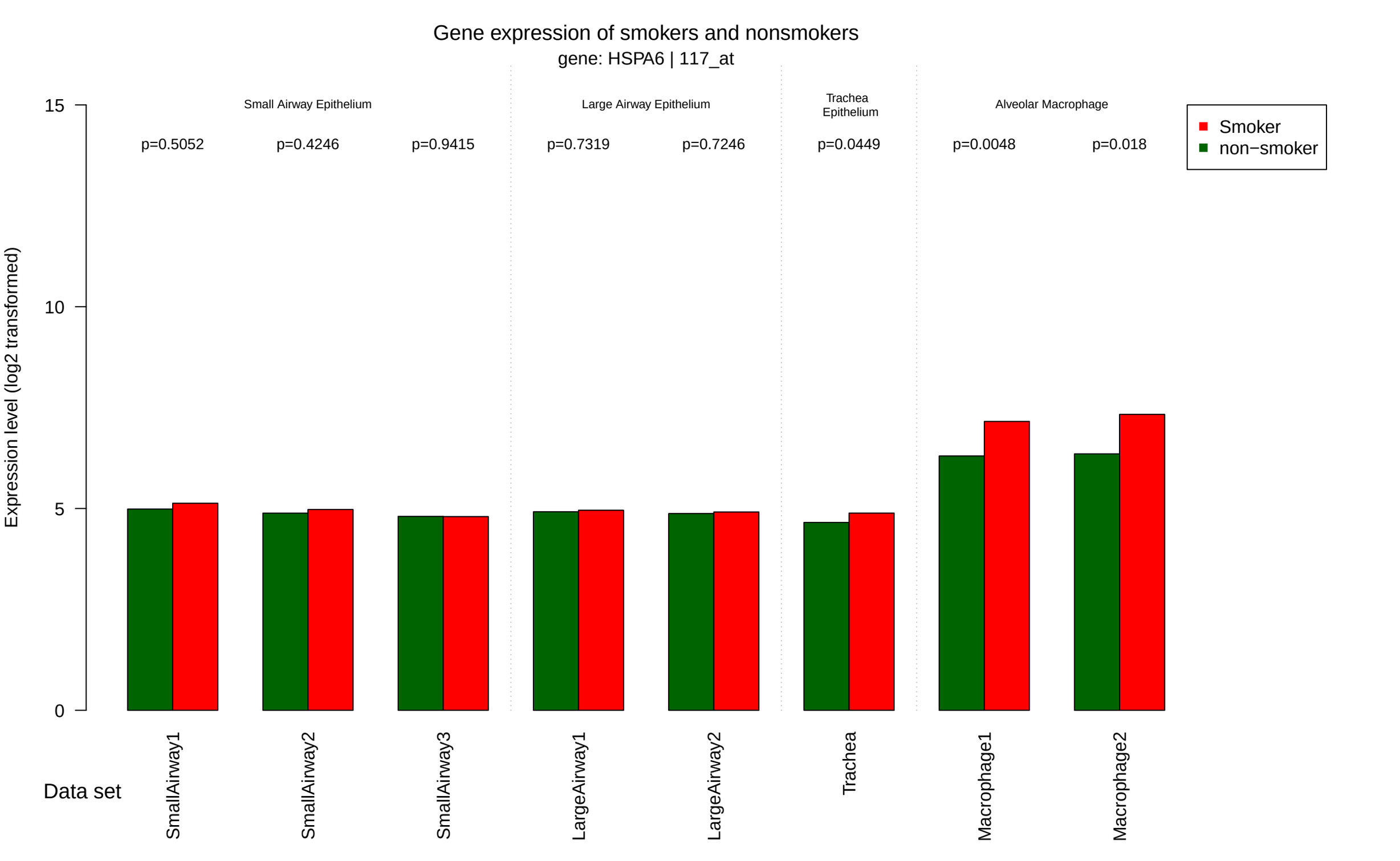
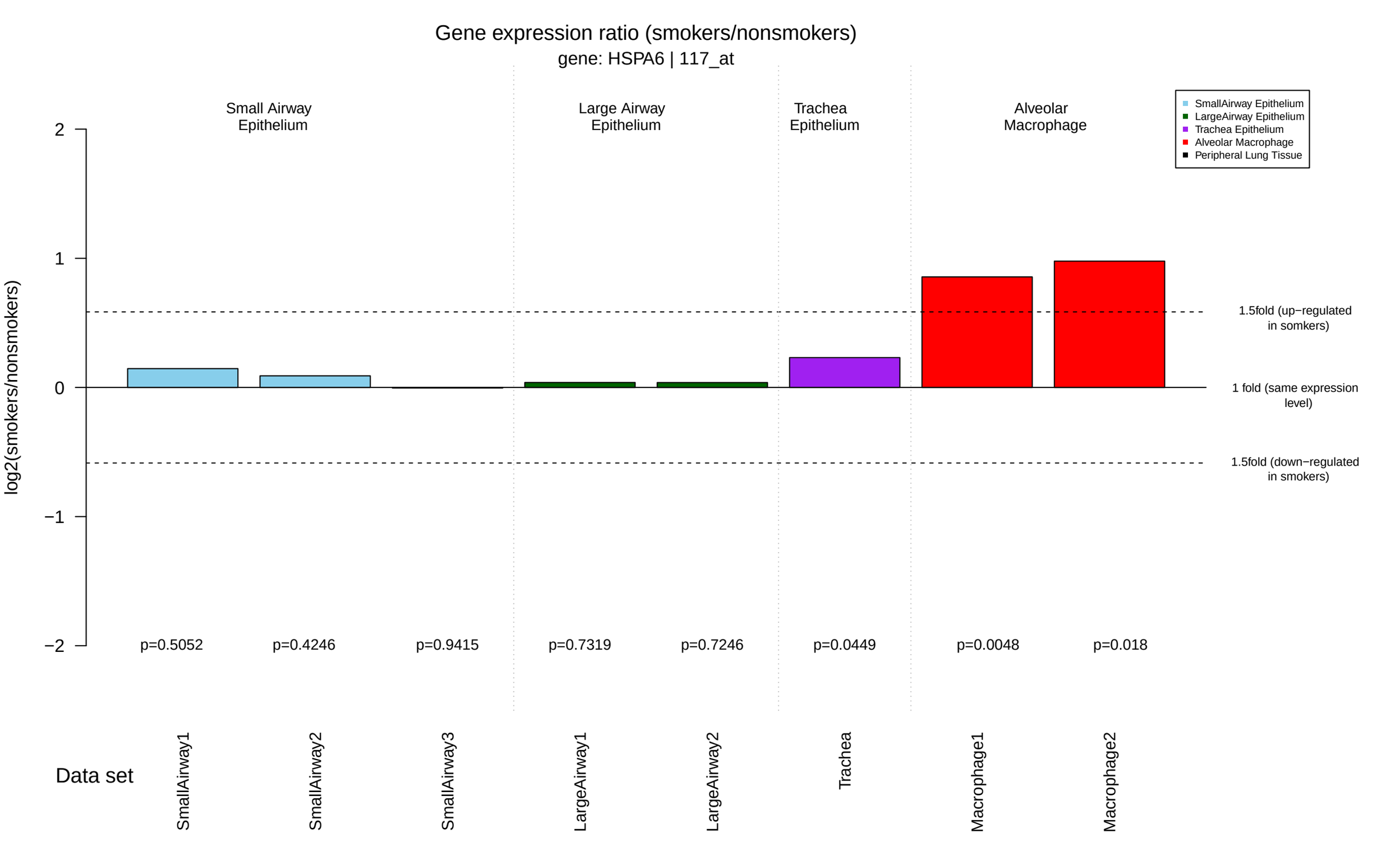
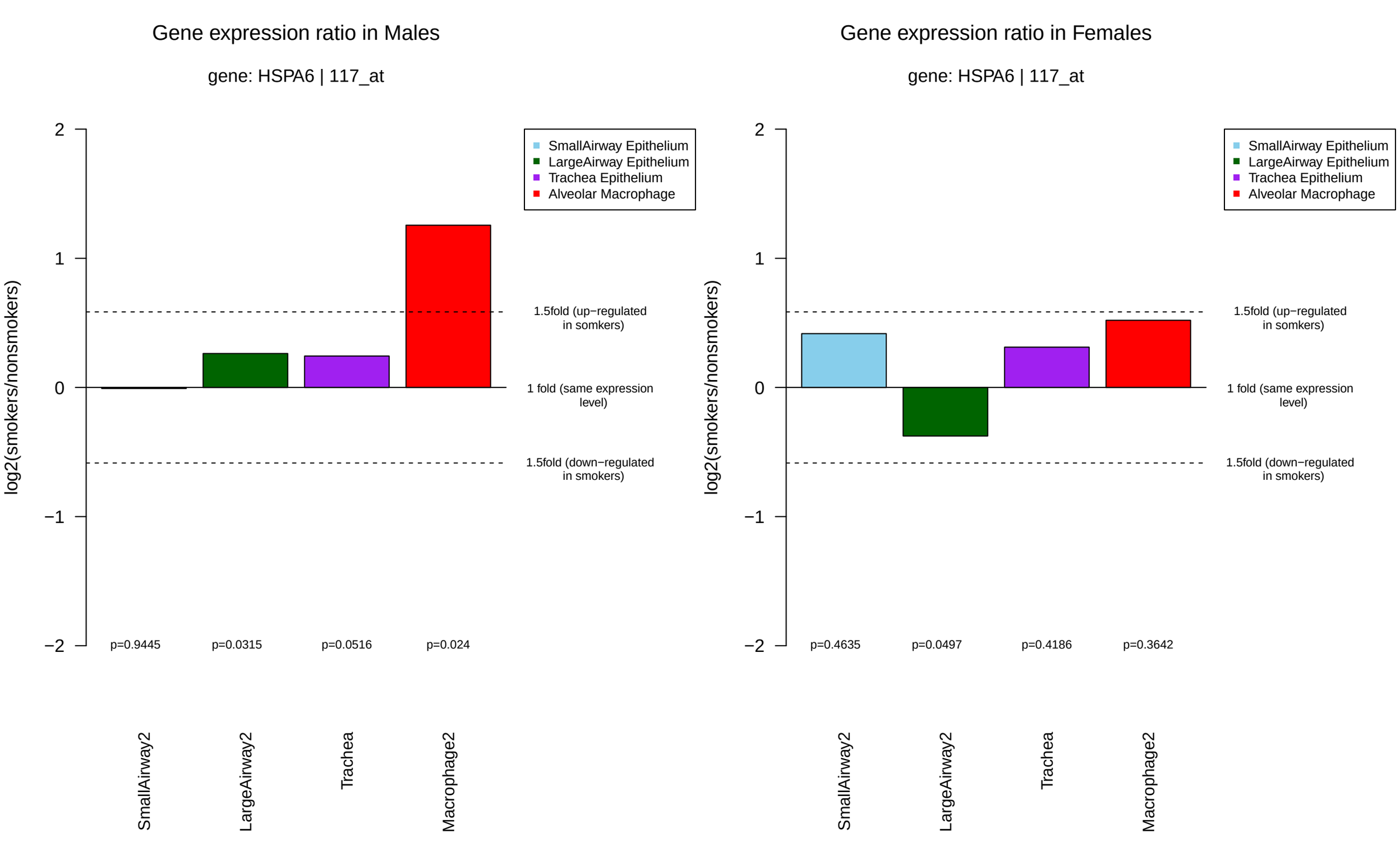
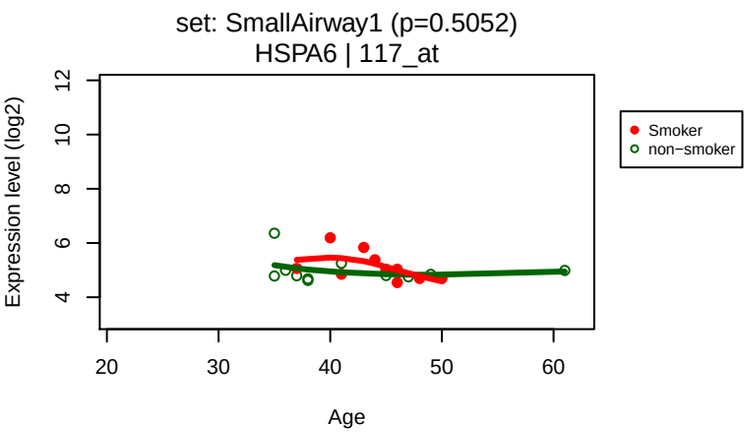
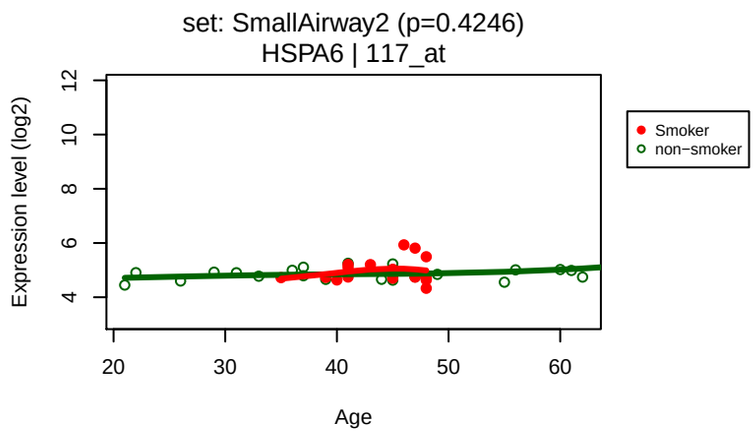
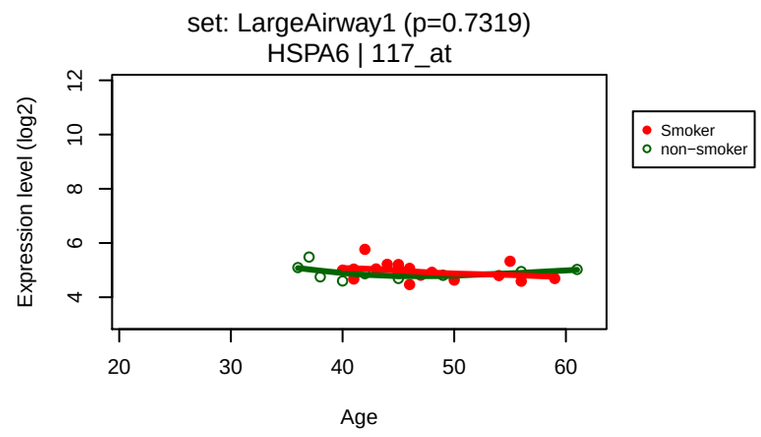

Gene expression of smokers and nonsmokers
gene: HSPA6 | 117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5052
p=0.4246
p=0.9415
p=0.7319
p=0.7246
p=0.0449
p=0.0048
p=0.018
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

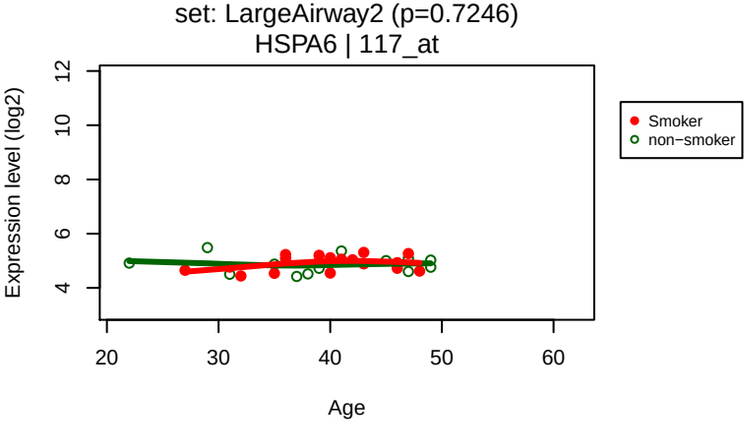

Gene expression ratio (smokers/nonsmokers)
gene: HSPA6 | 117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5052
p=0.4246
p=0.9415
p=0.7319
p=0.7246
p=0.0449
p=0.0048
p=0.018
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

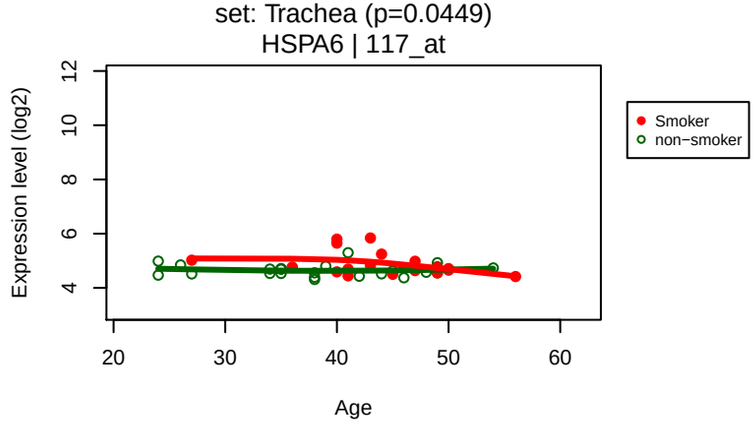

Gene expression ratio in Males
gene: HSPA6 | 117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9445
p=0.0315
p=0.0516
p=0.024
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA6 | 117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4635
p=0.0497
p=0.4186
p=0.3642
SmallAirway2
LargeAirway2
Trachea
Macrophage2

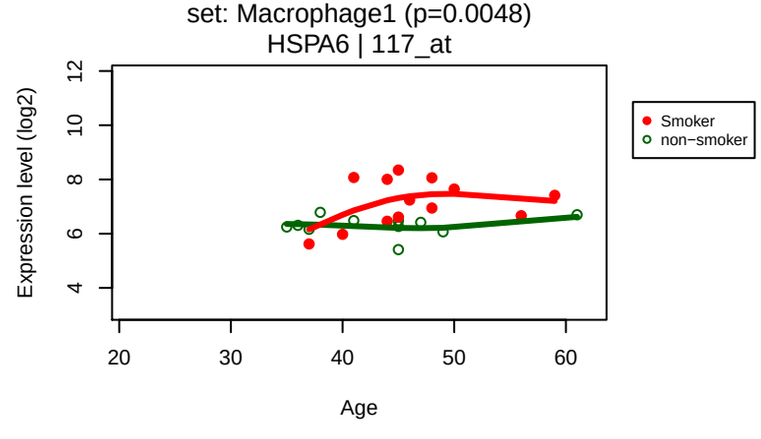

set: SmallAirway1 (p=0.5052)
HSPA6 | 117_at
Expression level (log2)
Age
Smoker
non−smoker

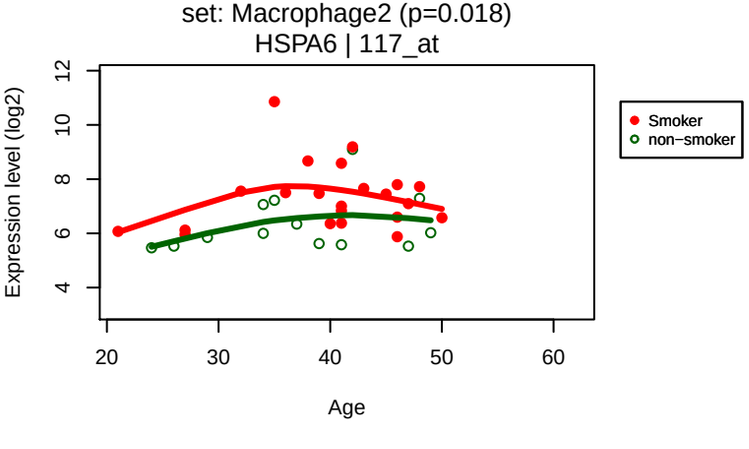

set: SmallAirway2 (p=0.4246)
HSPA6 | 117_at
Expression level (log2)
Age
Smoker
non−smoker

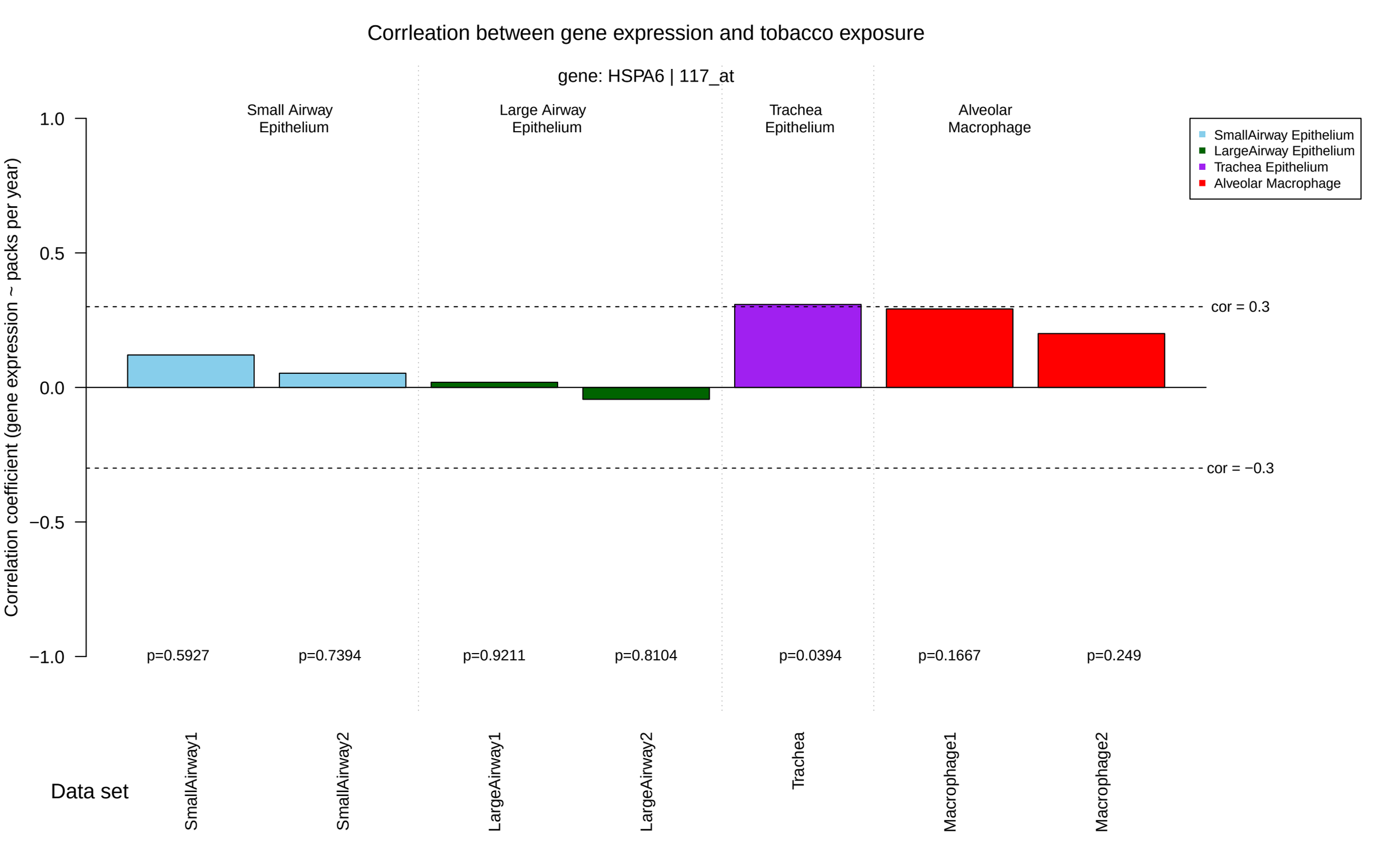

set: LargeAirway1 (p=0.7319)
HSPA6 | 117_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7246)
HSPA6 | 117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0449)
HSPA6 | 117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0048)
HSPA6 | 117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.018)
HSPA6 | 117_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA6 | 117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5927
p=0.7394
p=0.9211
p=0.8104
p=0.0394
p=0.1667
p=0.249
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2